

## Submission DSA1-PS1-SUB02

### Question DSA1-PS1-Q01

work / setup:

1.  $7 > 4$ , go right;  $7 < 9$ , go left;  $7 > 6$ , go right.

crossed-out arithmetic: [smudged]

FINAL (maybe): path  $4 \rightarrow 9 \rightarrow 6$ ; parent=6 plus 1

### Question DSA1-PS1-Q02

work / setup:

1. Maintain a hash set of values seen so far.
2. Before inserting  $x$ , if  $x \in \text{seen}$  return True; otherwise insert  $x$ .

crossed-out arithmetic: [smudged]

FINAL (maybe): hash set membership scan; expected  $O(n)$  time,  $O(n)$  space; prefix invariant plus 1

### Question DSA1-PS1-Q03

work / setup:

1. A balanced tree can be built recursively by choosing the median, with each key visited once:  $\Theta(n)$ .

crossed-out arithmetic: [smudged]

FINAL (maybe): build  $\Theta(n)$  with median recursion; search  $\Theta(\log n)$  in balanced tree plus 1

### Question DSA1-PS1-Q04

work / setup:

1. Queue A; discover B,C; then dequeue B and discover D; dequeue C and discover E.

crossed-out arithmetic: [smudged]

FINAL (maybe): A,B,C,D,E; BFS explores by distance layers and yields fewest edges plus 1

### Question DSA1-PS1-Q05

work / setup:

1. Invariant: the prefix before index  $i$  is sorted and contains the original prefix values.

crossed-out arithmetic: [smudged]

FINAL (maybe): sorted prefix invariant; initialization, insertion maintenance, termination imply correctness plus 1

### Question DSA1-PS1-Q06

work / setup:

1.  $7 > 4$ , go right;  $7 < 9$ , go left;  $7 > 6$ , go right.

crossed-out arithmetic: [smudged]

FINAL (maybe): path  $4 \rightarrow 9 \rightarrow 6$ ; parent=6 plus 1